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Lesson Notes

MODULE 7 | LESSON 1

SHRINKING (THE PROBLEM OF) THE COVARIANCE MATRIX

Reading Time	30 minutes
Prior Knowledge	Traditional portfolio theory and utility theory.
Keywords	behavioral finance, bounded rationality, satisfice, prospect theory, cognitive biases, emotional biases.

In the last module, we discussed factor models in detail—from their construction through regressions to the dangers of their popularity. In this module, we look at several techniques used to improve the inputs to the mean-variance optimization, starting with the covariance matrix.

1. An Overview of the Readings

The first required reading provide a gentle introduction to the importance—and trickiness—of estimating the covariance matrix. We also get our first glimpse of the Ledoit-Wolf shrinkage estimator, as well as a straightforward Python implementation of the Ledoit-Wolf shrinkage (LWS) in action (van Hoecke and Margenot). The comments on volatility, covariance, and their relationship to modern portfolio theory may be quite familiar to you by now, but the authors formulate these important points succinctly. Review them carefully. Hopefully, you also recognize the Python code for simulating dependent variables with random noise. Similarly, performing and plotting a linear regression may seem trivial to you. If not, read through the code and make sure it makes sense to you. As always, a great way to gain intuition about how code works is to copy and paste the code into your own notebook and run it yourself. Then, modify elements of the code and see if your modifications have the results you expect.

The second required reading builds on the introduction of LWS from the first reading and provides an overview of other methods used to construct the optimal covariance matrix, including the oracle approximating shrinkage (OAS). Also described at a high level are two non-shrinkage approaches to estimating a robust covariance matrix:

1. The **GraphicalLasso** estimator, which is especially useful when the sample size is small, uses L1 regularization to enforce a sparse inverse matrix.

2. The **Minimum Covariance Determinant (MCD)** estimator selects non-outliers within the sample with which to estimate an initial covariance matrix, then reweights each of the observations based on their **Mahalanobis distance** from the other points in the distribution. (Think of the Mahalanobis distance as just a multi-dimensional generalization of Euclidian distance.) This plot gives you an idea of how the use of the MCD estimator might differ from a classical statistical distance that does not incorporate the MCD.

We will not focus on these two approaches to robustification, but you should be aware of them. At the very least, you might be impressed by the ingenuity (on multiple levels) of the MCD approach. On the other hand, the variety of settings in which we encounter familiar regularization techniques such as Lasso should alert you to their importance.

The third and fourth short readings compare the two approaches, LWS and OAS, using simulated data and functions from the Python library scikit-learn, which you have no doubt already used.

The fifth reading formally introduces oracle approximating shrinkage. Indeed, in this order of readings, the functional implementation of OAS is presented before the academic paper that introduced the approach to the world and demonstrated its logical development. The idea is that you have a general sense of what OAS is before sifting through the technical details and a rigorous proof. You won't need to reproduce this proof, but you should nevertheless be able to follow its logic and make sense of the linear algebra involved, even if you are not familiar with all the identities relied upon. This applies to the proofs in the appendix as well. It will help for you to recall the meaning of the Frobenius matrix norm, which is just the square root of the sum of the squared elements of the matrix.

Figure 1: Frobenius Matrix Norm Formula

$$\|A\|_F = \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2 \right)^{1/2}$$

Source: Dax, Achiya. "From Eigenvalues to Singular Values: A Review." *Advances in Pure Mathematics*, vol. 3 no. 9B, 2013, pp. 8–24. https://www.scirp.org/html/2-5300515_41122.htm.

2. Why Shrink?: The Unstable Covariance Matrix

By now you have read several complaints, both in lesson notes and the required readings, about the sensitivity of mean-variance optimization (MVO) to estimation errors and how even a very small difference in the covariance matrix can result in very different portfolios that are optimized with MVO. The source of this instability is actually the **precision matrix**, which is simply the inverted covariance matrix, and the problem is especially bad when the covariance matrix has a high **condition number**. The condition number of a matrix indicates how sensitive to perturbations the solution involving the matrix is. Such perturbations include even very small changes, such as rounding, that one would ordinarily discount as inconsequential. Consider also the impact of even a single outlier on the correlation and variance, and so the covariance, between two variables. And now consider how multiple outliers in a large covariance matrix might wreak havoc on the performance of the portfolio constructed from the inverse of such a matrix. So when we say a

covariance matrix is ill-conditioned, we mean that the portfolio resulting from MVO involving that matrix may be very far from the *optimal* portfolio we hoped for, due to the outsized impact of any error.

Now consider that we usually take the sample covariance matrix as our estimator for the true or population covariance matrix. We do this for good reason: It is an unbiased estimator, and in fact, it is the unbiased estimator with the least variance (in other words, it is the Maximum Likelihood Estimate, or MLE). However, just because the estimator is unbiased does not mean it is perfect. For example, since the sample covariance matrix consists of some finite number of samples, it will almost surely contain at least some estimation error. Even if we could use an infinitely large sample, we have little reason to believe that the true current variances and correlations have not changed over the course of time. In short, there is no escaping estimation error, which leads to instability and the suboptimal performance of our so-called optimal portfolio.

For this reason, researchers and practitioners need a way to make their portfolio optimization process robust to such small errors. Shrinkage is one solution and possibly the most popular in finance. The basic idea of shrinkage is reducing the elements of the covariance matrix, especially the covariance elements off the diagonal, as opposed to the variances along the diagonal.

When van Hoecke and Margenot apply one of the most popular approaches (LW shrinkage), they not only demonstrate the decreased error (calculated as the sum of absolute differences between the LWS and sample covariance matrices), but they also make the very credible claim that this improvement “translates into **decreased volatility** and **turnover** rate in our portfolio, and thus **increased returns** when using the shrunk covariance matrix.”

- There is “decreased volatility” because the improved correlation input to the MVO means that portfolio volatility is better minimized: As a simple example, imagine that due to an outlier, you believed there was a strong negative correlation between two securities, so your optimal portfolio incorporated these two securities assuming that they hedged each other’s risk. When this supposed negative correlation was revealed by the market to be an estimation error, it was too late: The actual volatility (probably to the downside) had already been realized in your profit/loss statement.
- There is “decreased turnover” (which, among other things, saves transaction costs) because the updates to the optimal portfolio are less sensitive to the small changes that have occurred in the sample covariance matrix since the last time it was calculated. Remember that small changes in an ill-conditioned covariance matrix have an outsized effect, so even if the new time-series window for the covariance matrix update looks very similar to the last one, any small change could mean a large change in allocations. That means a lot of transaction fees for all the buy and sell orders.
- “Increased returns” follows directly from the above two points: Better hedging means lower losses. Less turnover means fewer transaction costs. Of course, the Sharpe ratio benefits, in both the numerator and the denominator.

3. Moral Hazard and Model Risk: Prescient Warnings from Ledoit and Wolf

As indicated above, you will learn from the required readings the mechanics of LWS as well as how to apply it straightforwardly in Python. Since none of the papers by Ledoit and Wolf are required readings, what follows are a few noteworthy observations by Ledoit and Wolf:

With the first sentence of the abstract of their humorously titled paper, “Honey, I Shrank the Covariance Matrix,” Ledoit and Wolf announce their serious intentions: “The central message of this paper is that nobody should be using the sample covariance matrix for the purpose of portfolio optimization” (1). They explain the pros and cons of using the sample covariance matrix: “Its advantages are ease of computation and the property of being unbiased (i.e., its expected value is equal to the true covariance matrix). Its main disadvantage is the fact that it contains a lot of estimation error when the number of data points is of comparable or even smaller order than the number of individual stocks; this is the common situation in financial applications” (5). Again, nothing new here but the points nevertheless bear repeating because they do frame the fundamental issue for portfolio optimization.

With that said, what truly stands out in this paper is the authors’ very down-to-earth advice to practitioners about how best to confront this dilemma—and how *not* to confront it: Some companies “have proposed proprietary methods to generate covariance matrices that are advertised as better suited to mean-variance optimization than the sample covariance matrix. The drawbacks are that any manager using them establishes a costly and indefinite dependence on an external entity who does not share in any downside risk, and that their proprietary methods are not open for independent inspection and verification, so one can never be sure what is really going on behind the curtain” (2).

If you cringe at the idea of blindly relying on a third party’s model, then you learned an important lesson from the subprime crisis discussed in Financial Markets. To be sure, if you do not know the mechanics of the model you are using, then you are not in a place to assess or manage your model risk—the risk that the model you are using is inappropriate for your purposes.

The potential for moral hazard arises whenever a third party influences decision making but does not bear any risk of the downside related to those decisions. Most of the investors that relied exclusively on the credit rating agencies’ models to assess the value and risk of mortgage-backed securities, collateralized debt obligations, and the like, learned this lesson at great cost: The rating agencies were paid by the issuers to rate these securities, but the ratings agencies did not invest in them, nor was there any other feedback loop so that inappropriate ratings would have direct financial implications on the ratings agencies. For this reason, it is said that the ratings agencies did not have “skin in the game,” i.e., they had nothing to lose if they did a poor job of assessing the risk of these securities. (Indirectly, of course, the ratings agencies were indeed hurt by their own poor assessment performance: Their reputations were severely tarnished.) Interestingly, Ledoit and Wolf make this important point about moral hazard in their paper published in 2003, a few years before this bleak phenomenon came to dominate headlines and changed investors’ minds about outsourcing their most important modeling responsibilities.

4. The Rao-Blackwell Theorem and the Sufficient Statistic

Given the prominence of the Rao-Blackwell (RB) theorem in the Rao-Blackwell estimator and the OAS that builds on it, it is worth our time to expand on the definition of the RB theorem provided in the article and review the properties of the sufficient statistic. As described by Hardy, “a **sufficient**

statistic for a statistical model is a statistic that captures all information relevant to statistical inference within the context of the model" (Scholarpedia). Such a statistic is considered sufficient because no other information is needed, and so a model that relies on such a statistic is likely more efficient and better because it does not need to incorporate additional data.

Now that we remember the definition of a sufficient statistic, let us get some clarity about the Rao-Blackwell Theorem before you encounter the exact statistical definition in the article by Chen et al. According to Scholarpedia, the "**Rao-Blackwell Theorem** provides a process by which a possible improvement in efficiency of an estimator can be obtained by taking its conditional expectation with respect to a sufficient statistic." Technically, "the precise version of the theorem only says that this improved estimator is no worse than the estimator that was to be improved. In practical applications, however, it is enormously better" (Hardy).

Chen et al. exploit this increase in efficiency and do appear to make significant improvements with their extension of the LW model, but there is a price to be paid for the improvement: As they make very clear, when the Gaussian distribution can be assumed, the sample covariance is a sufficient statistic for their model; LW do not make this assumption so their model requires additional information and the sample covariance is not a sufficient statistic in that case. Is it surprising that if we are allowed to make a distributional assumption, such as that our samples are Gaussian, we can reduce the MSE? And that such an approach provably dominates when the simulated data are again constructed to be Gaussian? The implied caveat is to ensure that your data is Gaussian—or at least Gaussian *enough*—so that this assumption does not backfire.

5. Conclusion

In this lesson, we discussed the unstable matrix problem in more detail, as well as robustification techniques to help solve this problem. In this vein, we briefly defined the GraphicalLasso and Minimum Covariance Determinant (MCD) estimators and showed the associated SciKit-learn methods. Then, we discussed the concept of shrinkage. The process of shrinking a covariance matrix to make it more robust requires a shrinkage coefficient, so we reviewed some approaches to finding the optimal shrinkage coefficient: Ledoit-Wolf, Rao-Blackwell Ledoit-Wolf, and the Oracle-Approximating Shrinkage.

In the rest of this module, we continue our work on the covariance matrix, finding ways to modify it so that the optimization procedures that rely on it produce better performing portfolios. The next lesson will focus on removing the noise (via de-noising) and otherwise improving the signal in the covariance matrix (via de-toning).

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